

B.Tech III Year II Semester (R19) Supplementary Examinations January 2024

INTRODUCTION TO MACHINE LEARNING

(Common to CSE and IT)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- (a) State Version Space representation theorem. 2M
 - (b) Define Entropy and Information Gain. 2M
 - (c) What factors contribute to the popularity of Genetic Algorithm? 2M
 - (d) What is the role of the activation function in a Neural Network? 2M
 - (e) State Bayes Theorem. 2M
 - (f) Define Maximum Likelihood Estimation. 2M
 - (g) What is Regression and mention its types? 2M
 - (h) Write the applications of Radial Basis Function Network. 2M
 - (i) Define Reinforcement Learning. 2M
 - (j) Mention two difference between Induction and Deduction. 2M

PART – B
(Answer all the questions: 05 X 10 = 50 Marks)

- 2 (a) Define Concept Learning. Explain the task of Concept Learning. 5M
- (b) What is Inductive Bias? Discuss with respect to decision tree learning. 5M

OR

- 3 Construct a decision tree using ID3 considering the following training examples for target function Play Tennis. 10M

Day	Outlook	Temperature	Humidity	Wind	Play
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

- 4 (a) What is Perceptron? Explain the representational power of Perceptron. 5M
- (b) Discuss Genetic Programming in detail. 5M

OR

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- 5 Explain Multilayer Neural Network with back propagation in detail. 10M
- 6 (a) What are Bayesian Belief network? Where are they used? 5M
(b) Summarize Naïve Bayes Classifier with an Example. 5M
- OR**
- 7 (a) Discuss Minimum Description Length Principle in brief. 5M
(b) Write a short note on EM Algorithm. 5M
- 8 (a) Define the following terms with respect to K - Nearest Neighbour Learning: 5M
(i) Regression, (ii) Residual, (iii) Kernel Function.
(b) Demonstrate Locally Weighted Linear Regression. 5M
- OR**
- 9 Explain Case based learning in detail. 10M
- 10 (a) Illustrate FOCL algorithm of learning first order rules. 5M
(b) Discuss the learning tasks and Q-learning in the context of reinforcement learning. 5M
- OR**
- 11 (a) Define Inverse Resolution? Explain Inverse Resolution in First Order Logic. 5M
(b) What is Explanation Based Learning? Give an example. 5M
